PhD Lecture in Behavioral Asset Pricing: Limits to Arbitrage

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Overview

- *Recap (Lecture 1)*: Return anomalies persist even though these are all simple strategies using public information
- *Today*: Why do rational arbitrageurs not fully correct these anomalies?
- Applications of behavioral finance to asset pricing posit that irrational investors affect prices
- There is a classic critique of this idea
  - The "arbitrage critique"
- According to this critique, irrational investors cannot affect prices for any significant amount of time
  - As soon as irrational investors move prices, this creates an attractive opportunity for rational investors
  - The rational investors trade against the mispricing, quickly eliminating it ("arbitrage")
- A major achievement of behavioral finance is to successfully push back against the arbitrage critique
  - i.e. to show that there are "limits to arbitrage"

Overview, ctd.

- **Terminology:**
  - The "fundamental value" of an asset is its price in an economy with rational investors and no frictions
    - The price that properly reflects all available public information
    - The efficient market price
  - In an economy with frictions, or where agents are not fully rational, an asset's price can depart from fundamental value
    - This is a "mispricing"
    - Or an "inefficiency"
  - Rational investors are sometimes referred to as "arbitrageurs"
  - Less than fully rational investors are sometimes referred to as "noise traders"

Overview, ctd.

- **Arbitrage**
  - An investment strategy that guarantees a positive payoff in some state, with no possibility of a negative payoff, no net investment, and at arbitrary scale
- $\Rightarrow$ Money Machine
- Arbitrage opportunities are inconsistent with common notions of equilibrium
  - If arbitrage opportunities exist, someone will act and prices will change, so this is not a stable situation
- **Law of one price**
  - Two assets with the same payoff in every state must have the same price
  - Key to the law of one price is perfect substitutes (exactly the same payoffs)
  - Underlies much of the design of derivatives

Arbitrage, Formally

An arbitrage is a portfolio with payoffs V_s across states s satisfying:

$$V_0 \leq 0, \quad V_s \geq 0 \ \ \forall s, \quad \Pr[V_s > 0] > 0$$

- $V_0 \leq 0$: zero (or negative) net investment today
- $V_s \geq 0$ for all s: no possibility of a loss in any future state
- $\Pr[V_s > 0] > 0$: strictly positive probability of a gain in some state

Limits to Arbitrage: Theory

- What is the response to the arbitrage critique?
  - The arbitrage critique says that it will be easy for rational investors to correct a mispricing
  - In reality, however, it is not easy
    - There are risks and costs that limit arbitrageurs' ability to correct a mispricing
    - This allows irrational investors to affect prices significantly and for a long time
  - Specific limits to arbitrage:
    - **Risks:** fundamental risk, noise trader risk, horizon risk, model risk
    - **Costs:** trading costs, shorting costs, learning and information-processing costs

Limits to Arbitrage: Theory, ctd.

- **Fundamental risk**
  - The risk that there will be adverse news about the fundamental value of the mispriced asset
  - Usually a problem because perfect substitutes are not available for what you really want to trade on
    - E.g., suppose you think Ford is underpriced; your best hedge against unexpected changes is GM
    - So you long Ford and short GM
    - This strategy is immune to auto industry shocks common to both Ford and GM
    - However, this is not risk-free arbitrage as you are exposed to the risk that Ford's fundamentals may further deteriorate relative to GM's

Limits to Arbitrage: Why Firm-Specific Risk Cannot Be Hedged

1. Return decomposition: industry exposure + firm-specific risk

For firm i and its same-industry peer j:

$$r_{it} = \alpha_i + \underbrace{\beta_i r_{I,t}}_{\text{industry exposure}} + \underbrace{\varepsilon_{it}}_{\text{firm-specific}}, \qquad r_{jt} = \alpha_j + \beta_j r_{I,t} + \varepsilon_{jt},$$

where $\mathrm{Corr}(\varepsilon_{it}, \varepsilon_{jt}) = \rho_{ij}$ measures how close a substitute j is for i.

2. Arbitrage portfolio: industry factor cancels

With i overvalued ($\theta_{it} > 0$), go long j, short i:

$$r_t^{arb} = r_{jt} - \frac{\beta_j}{\beta_i} r_{it} = \left(\alpha_j - \tfrac{\beta_j}{\beta_i}\alpha_i\right) + \underbrace{\left(\varepsilon_{jt} - \tfrac{\beta_j}{\beta_i}\varepsilon_{it}\right)}_{\text{unhedgeable firm-specific risk}}.$$

Residual variance: with $\beta_i = \beta_j$ and $\sigma_{\varepsilon_i} = \sigma_{\varepsilon_j} = \sigma_\varepsilon$,

$$\boxed{\sigma^2_{arb} = 2\,\sigma^2_\varepsilon\,(1 - \rho_{ij})}$$

Perfect substitute: $\rho_{ij} \to 1 \Rightarrow \sigma^2_{arb} \to 0$ vs. No close substitute: $\rho_{ij} \ll 1 \Rightarrow \sigma^2_{arb} \approx 2\sigma^2_\varepsilon$

3. Risk aversion bounds the position

$$x^* = \frac{\theta_{it}}{\gamma\,\sigma^2_{arb}} = \frac{\theta_{it}}{2\gamma\,\sigma^2_\varepsilon(1-\rho_{ij})}, \qquad \frac{\partial x^*}{\partial \rho_{ij}} > 0, \quad \frac{\partial x^*}{\partial \sigma^2_\varepsilon} < 0$$

Takeaway: No close substitute ⇒ fundamental risk remains ⇒ positions bounded ⇒ mispricing θ_{it} persists.

Limits to Arbitrage: Theory, ctd.

- **Noise trader risk** (De Long et al., 1990; Shleifer and Vishny, 1997)
  - The risk that, as a result of the mispricing worsening in the short run, the arbitrageur is forced to close out their trade prematurely at a loss
  - This risk arises because real-world arbitrageurs typically manage other people's money and because they use leverage
- **An example:**
  - Arbitrageurs engage in arbitrage in reaction to a stock mispricing
  - Mispricing persists for an extended period
  - Clients deem the arbitrageur incompetent and demand withdrawal of their funds
  - To deliver cash, the manager must unwind the position at a loss
  - This threat causes managers to be less willing to take advantage of these opportunities, exacerbating the problem of pricing inefficiency

Limits to Arbitrage: Theory, ctd.

- Hedge funds' use of leverage and short selling compounds the problem
  - If a fund borrows money to buy an undervalued asset and the mispricing then worsens
  - The lender, seeing the value of their collateral decline, may call the loan
  - This forces the fund to close out its trade at a loss
  - Similarly, if the fund shorts an overvalued asset and the asset then goes up in price
  - The lender will demand additional margin
  - If the fund cannot meet this demand, the short position will be closed out at a loss
  - A fund manager who foresees these potential outcomes will trade less aggressively ex ante against the mispricing
- Real-world case: the failure of LTCM

Case Study: Long-Term Capital Management (1994–1998)

The fund.

- Founded 1994 by John Meriwether; principals included Myron Scholes and Robert Merton (Nobel 1997).
- Strategy: *convergence trades* — long cheap securities, short rich ones, betting spreads revert: swap spreads, Treasury on-the-run/off-the-run, merger arbitrage, Russia/Emerging-market debt.
- Peak equity: $\approx$ \$4.8bn (early 1998); balance sheet $>$ \$100bn; off-balance-sheet derivatives notional in the *trillions*. Leverage: $> 25\times$ (far higher counting derivatives).
- Positions were small per-trade, but massively correlated: all were short-liquidity, short-volatility bets on spread convergence.

The collapse (Aug–Sep 1998).

- Trigger: Russia defaults (Aug 17, 1998) $\Rightarrow$ global flight to liquidity and quality.
- Spreads *diverge* instead of converging; LTCM's losses $\approx$ \$4.5bn in weeks.
- Banks raise margin requirements $\Rightarrow$ forced deleveraging into falling markets (liquidity spiral).
- Sept 23, 1998: Fed brokers a \$3.6bn private recapitalization by 14 banks — fund deemed too interconnected to fail.

LTCM Through the Lens of Limits to Arbitrage

LTCM's trades were *convergence* trades — each had a substitute, but **no perfect one**.

1. Fundamental risk: imperfect substitutes

On-the-run vs. off-the-run Treasuries differ in liquidity, not just price:

$$\sigma^2_{arb} = 2\,\sigma^2_{\varepsilon}(1-\rho_{ij}) > 0$$

In a crisis, $\rho_{ij} \to 0$: the "substitute" stops hedging exactly when it matters most.

2. Noise-trader / wealth risk

Sentiment shocks (flight to quality) moved prices *further* from fair value before convergence:

$$x^* = \frac{\theta}{\gamma\,\sigma^2_{arb}} \;\to\; 0 \quad \text{as } \sigma^2_{arb} \uparrow$$

LTCM's own capital shrank with losses $\Rightarrow$ forced unwinding at the worst moment.

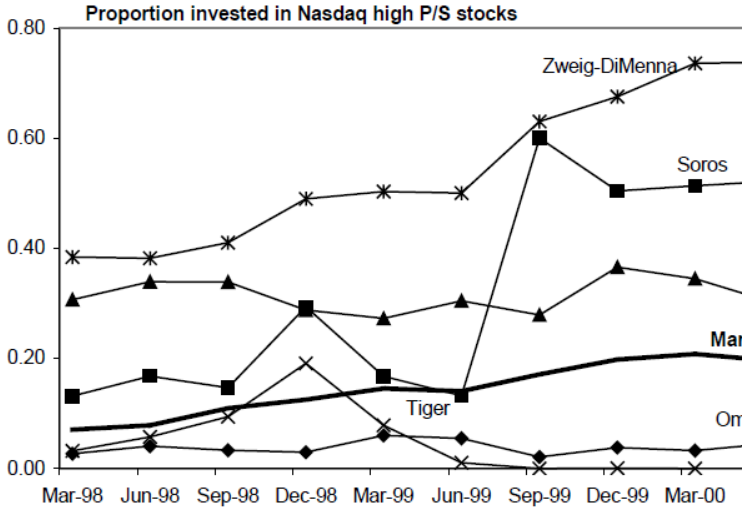
3. Margin constraints & liquidity spiral

$$E_t[r_j] = R_f + \gamma\,\sigma_j^2\,x_j + \mu\,m_j$$

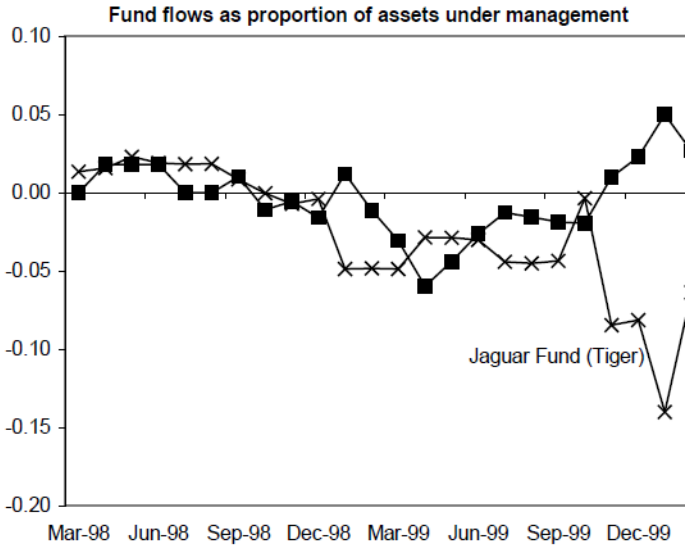
As margins m_j rose, $\mu > 0 \Rightarrow$ forced fire sales $\Rightarrow$ spreads widen further $\Rightarrow$ deeper losses: a *feedback loop*.

Lesson: Even a "riskless" arbitrage is exposed to **(i)** substitute imperfection, **(ii)** funding/wealth risk, **(iii)** margin spirals. Leverage amplifies all three. *The market can stay irrational longer than LTCM could stay liquid.*

Arbitrage in Reality



Arbitrage in Reality, ctd.



Limits to Arbitrage: Theory, ctd.

- **Trading costs**
  - Brokerage commissions
  - Bid/ask spreads
  - Price impact
- **Shorting costs**
  - Short selling involves borrowing shares from a lender, and the seller needs to pay lending fees
  - The lending fees could be very high for certain stocks in certain periods
  - Short selling is occasionally banned by regulators
  - Some countries do not allow short selling at all
- **Learning costs**
  - Arbitrageurs need to first be aware of the existence of anomalies
  - McLean and Pontiff (JF 2016) show that academic publication of anomalies helps disseminate knowledge of the anomalies and reduce their profitability
- **Information processing costs**
  - Even though arbitrageurs are aware of the anomalies, they need to process necessary data at low cost in order to trade on it

Limits to Arbitrage: Theory, ctd.

- Thus far, we have examined the "static" version of the arbitrage critique
- There is also a dynamic version, which says:
  - Even if rational traders are unable to quickly correct a mispricing, assets will eventually become correctly priced
  - By trading sub-optimally, irrational traders become poorer over time
  - As a result, they eventually become irrelevant to the pricing of assets
- This idea has generated a sizable literature on the "survival of noise traders"
- How compelling is the dynamic arbitrage critique?
  - It may take a very long time for irrational traders to exit the market (longer than most arbitrageurs' performance evaluation horizons)
  - Noise traders typically have other sources of income (labor income) to subsidize their investment losses
  - A new generation of irrational investors enters the market

Limits to Arbitrage: Evidence

- There are good theoretical reasons why arbitrage may be limited
  - But is there evidence that arbitrage is limited?
- Any example of a substantial, long-lived mispricing would be evidence of limits to arbitrage
  - But such examples are not easy to document
- Still, there are some pricing patterns that are widely viewed as mispricings
  - These constitute evidence of limits to arbitrage
  - And highlight the role played by each type of friction
- Specifically, we look at:
  - Twin shares
  - Equity carve-outs
  - Index inclusions
  - Anomalies

Twin Shares

- Twin shares are distinct shares that are claims on the same cash flows
- In 1907, Royal Dutch and Shell merged their interests while remaining separate entities
  - Royal Dutch has a claim to 60% of the combined firm's cash flows
  - Shell has a claim to 40% of the combined firm's cash flows
- In an economy with rational investors and no frictions, the market value of Royal Dutch should equal 1.5 times that of Shell:

$$MV_{RD} = 1.5 \times MV_S$$

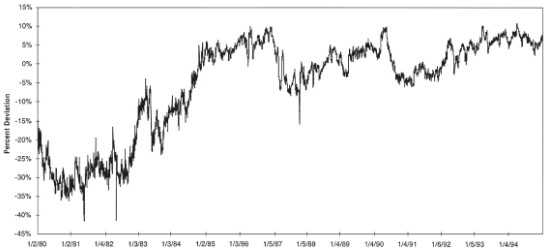


Fig. 1. Log deviations from Royal Dutch/Shell parity. Source: Froot and Dabora (1999).

Long-lived mispricing: hard to correct.

Twin Shares, ctd.

- **Example: AH Premium**
  - Dozens of Chinese companies' stocks are listed in both China's domestic stock market (A shares) and the Hong Kong stock market (H shares)
  - A shares usually trade at a persistent premium over H shares of the same company



AH premium, Chinese A vs. H shares

Equity Carve-Outs

- Mitchell, Pulvino, and Stafford (2002) study equity carve-outs between 1985 and 2000
  - Cases where a parent company conducts an IPO for shares in a subsidiary firm and retains more than 80% ownership
- They find 70 cases where $MV_{Parent} < 0.8 \times MV_{Subsidiary}$
  - Implies that parents' stub assets are negative
- Why is this mispricing hard to correct?
  - Fundamental risk (in 30% of cases, there is adverse fundamental news)
  - Noise trader risk
  - Shorting costs

Equity Carve-Outs, ctd.

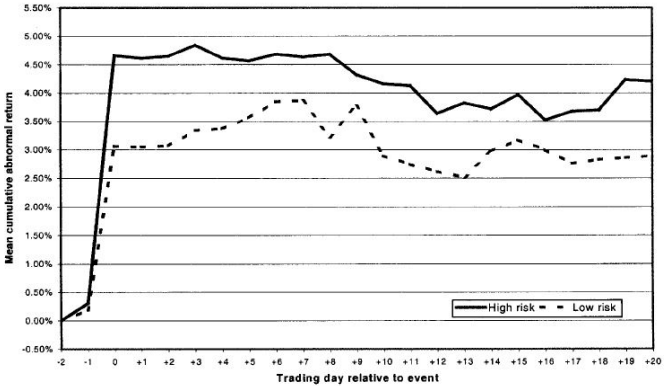
- **Palm/3Com** (March 2000)
  - 3Com sells 5% of Palm in an IPO; it will spin off the remaining 95% within 9 months
  - Each 3Com share is a claim to 1.5 shares of Palm post-spinoff
- **Worked arithmetic:**
  - $P_{Palm} = \$95$ (observed)
  - Lower bound on P_{3Com}: $1.5 \times \$95 = \$142.50$
  - $P_{3Com} = \$81$ (observed)
  - Implied stub value: $P_{3Com} - 1.5 \times P_{Palm} = 81 - 142.50 = -\$61.50$
- **Discussion:** The market values 3Com's non-Palm assets at $< \$0$. Why does an arbitrageur not short 3Com and buy Palm?
- **Limits at work:** very few Palm shares available to short

Index Inclusions

- Stocks are regularly added to the S&P 500 index to replace other stocks that have been removed due to mergers or bankruptcies
- **Facts:** on both the announcement and execution days, the included stocks jump in price by 3.5% on average, and the jump is not reversed
  - Harris and Gurel (1986), Shleifer (1986)
- The price jump is widely viewed as a mispricing
  - S&P's goal is to create a representative index, not to send signals about the included stocks' investment value
  - As such, the effect is evidence that there are limits to arbitrage
- To correct the mispricing, an arbitrageur would short the included stocks and buy substitute stocks
  - What are the limits to arbitrage in this case?

Index Inclusions, ctd.

- Wurgler and Zhuravskaya (2002) show that included stocks with worse substitutes (higher idiosyncratic risk) jump up more upon inclusion



Wurgler and Zhuravskaya (2002)

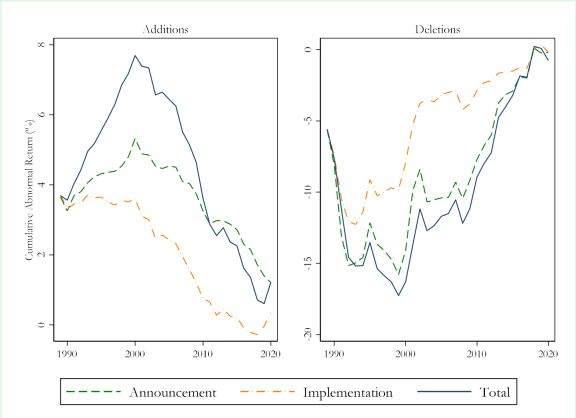
The Disappearing Index Effect

- The abnormal return associated with a stock being added to the S&P 500 has fallen from an average of 7.4% in the 1990s to less than 1% over the past decade (Greenwood and Sammon, 2025)
- This has occurred despite a significant increase in the share of stock market assets linked to the index
- The main reason is that professional active investors provide liquidity to accommodate the demand of passive indexers

The Disappearing Index Effect, ctd.

Figure 2. Average index effect by year.

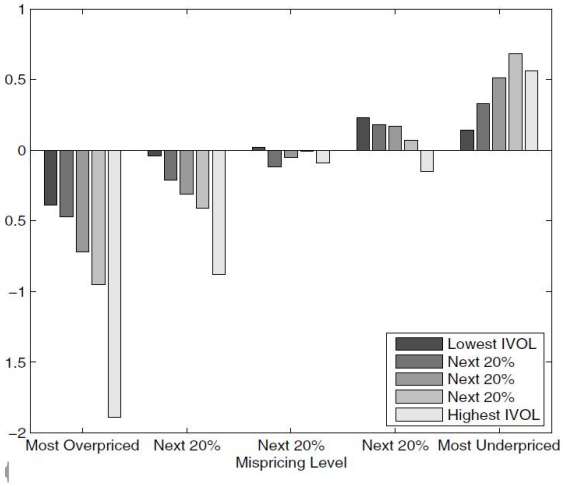
For each event, we compute the cumulative abnormal return over the period of interest. Lines represent 10-year moving averages of the yearly averages.



Greenwood and Sammon (JF 2025)

Anomalies

- If anomalies are the result of mispricing, they should be more pronounced among stocks with more severe limits to arbitrage
  - Using institutional ownership as a proxy for short-sale constraints, Nagel (2005) finds that the short leg of anomalies is stronger among stocks with lower institutional ownership
  - Using idiosyncratic volatility as a proxy for arbitrage risk, Stambaugh, Yu, and Yuan (2015) find that mispricing is amplified among stocks with the highest idiosyncratic volatility



Stambaugh, Yu, and Yuan (2015)

The Causal Effects of Limits to Arbitrage

- Chu, Hirshleifer, and Ma (2020) show the causal effects of limits to arbitrage on asset pricing anomalies
- They use Regulation SHO as a quasi-experiment:
  - The SEC removes short-sale restrictions on a randomly selected group of stocks (pilot stocks) within the Russell 3000 index
  - The pilot program period is from May 2, 2005 to August 6, 2007
- The research design is a standard difference-in-differences test:
  - Construct anomaly long-short portfolios for pilot stocks and non-pilot stocks separately
  - Compare the return spread of the long-short portfolios for pilot firms relative to non-pilot firms, before and during the pilot period
- Their main findings are:
  - Anomalies become weaker for pilot stocks relative to non-pilot stocks during the pilot period
  - The effect comes mostly from the short-leg portfolio

The Causal Effects of Limits to Arbitrage, ctd.

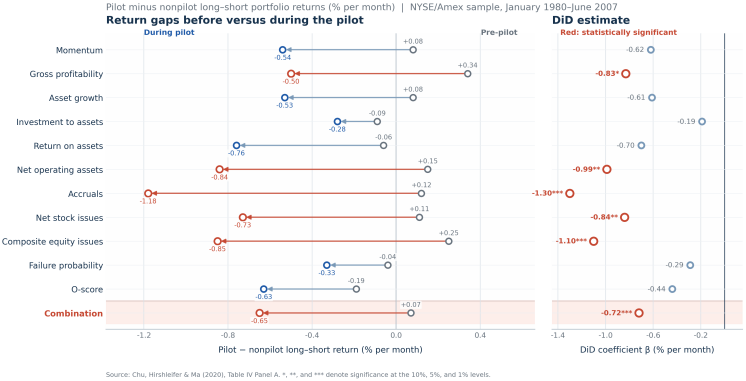
The main difference-in-differences test employs the following specification:

$$r_{it} = \alpha_t + \beta_1 Pilot_i + \beta\, Pilot_i \times During_t + \epsilon_{it}, \tag{1}$$

where r_{it} is the monthly return of portfolio i, which can be the long leg, the short leg, or the long-short portfolio of an anomaly, in month t; α_t denotes month fixed effects, which absorb the main effect of $During_t$; $Pilot_i$ is a dummy variable which is equal to one if portfolio i is constructed from pilot stocks, and zero otherwise; $During_t$ is a dummy variable equal to one if month t falls between June 2005 and July 2007, i.e. the full months spanned by the Regulation SHO pilot program (May 2, 2005 to August 6, 2007). The coefficient of interest is β on the interaction term.

The Causal Effects of Limits to Arbitrage, ctd.

Table IV Panel A — Relaxing short-sale constraints weakened long-short anomaly returns

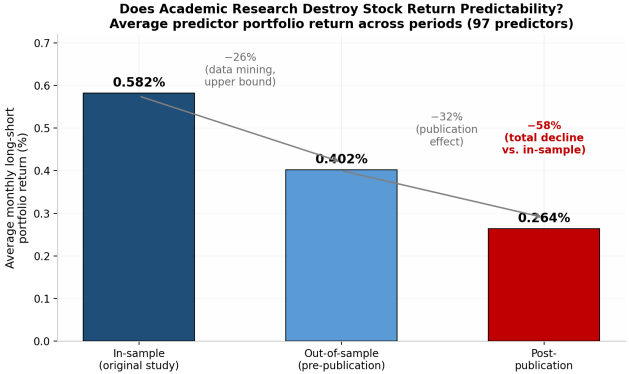


Source: Chu, Hirshleifer & Ma (2020), Table IV Panel A. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Chu, Hirshleifer, and Ma (JF 2020)

Learning Costs as a Limit to Arbitrage

- McLean and Pontiff (JF 2016) report the average monthly long-short returns of the 97 predictor portfolios across three periods: 0.582% in-sample $\rightarrow$ 0.402% out-of-sample (pre-publication) $\rightarrow$ 0.264% post-publication.



Information Processing Costs as a Limit to Arbitrage

- Bowles, Reed, Ringgenberg, and Thornock (2024)
- Anomaly returns are concentrated in the first month after anomaly-relevant information is released, and the returns decay quickly thereafter

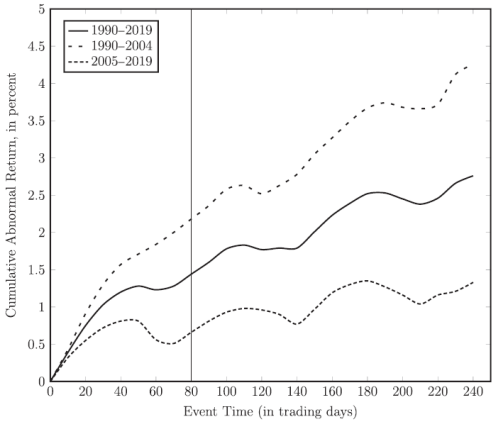
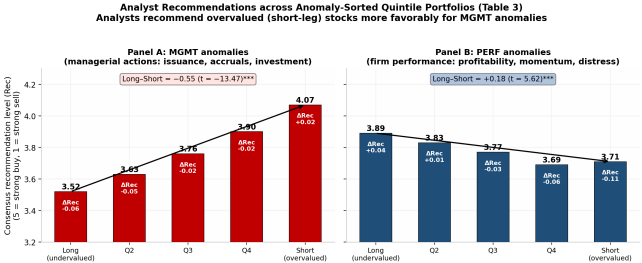


Figure 1. Anomaly returns in event time. The figure shows the profile of abnormal returns for the average anomaly (equally weighted) in event time. The figure shows the returns across the entire sample (1990 to 2019), across the early period of our sample (1990 to 2004), and across the recent years of our sample (2005 to 2019). The solid vertical bar at $t = 80$ trading days indicates when the FF92 rebalancing date occurs, on average.

My Research: Analyst Bias as a Limit to Arbitrage

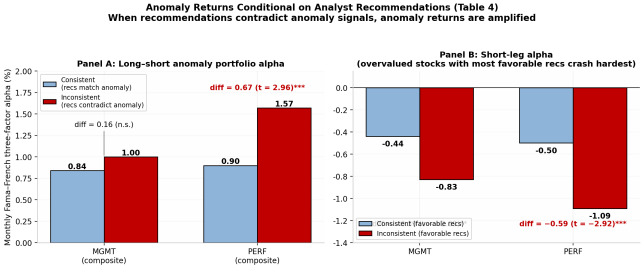
- **Li, Guo, and Wei (JFE 2020)**
- Consensus recommendations are more favorable for stocks classified as overvalued, and these stocks subsequently underperform significantly



Li, Guo, and Wei (JFE 2020)

My Research: Analyst Bias as a Limit to Arbitrage, ctd.

- **Li, Guo, and Wei (JFE 2020)**
- When recommendations contradict anomaly signals, anomaly returns are amplified
- **Implication:** Biased recommendations are themselves a friction that impedes the correction of anomalies



Li, Guo, and Wei (JFE 2020)

Summary

- The literature on limits to arbitrage has been influential
  - There is now wide agreement among academics (and practitioners) that arbitrage is limited
  - Although there is some disagreement as to how limited it is
- This success was one reason why behavioral finance "took off" in the 1990s
- Still, we should not be complacent
  - Whenever we argue that irrational investors' demand affects prices, we should ask if arbitrage is truly limited
- **Discussion:** With new technologies that expedite information collection and processing, which friction do you expect to matter less and what will remain?
- *Next (Lecture 3)*: What are the sources of these noise-trader demands?